JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO UVANCHITRA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

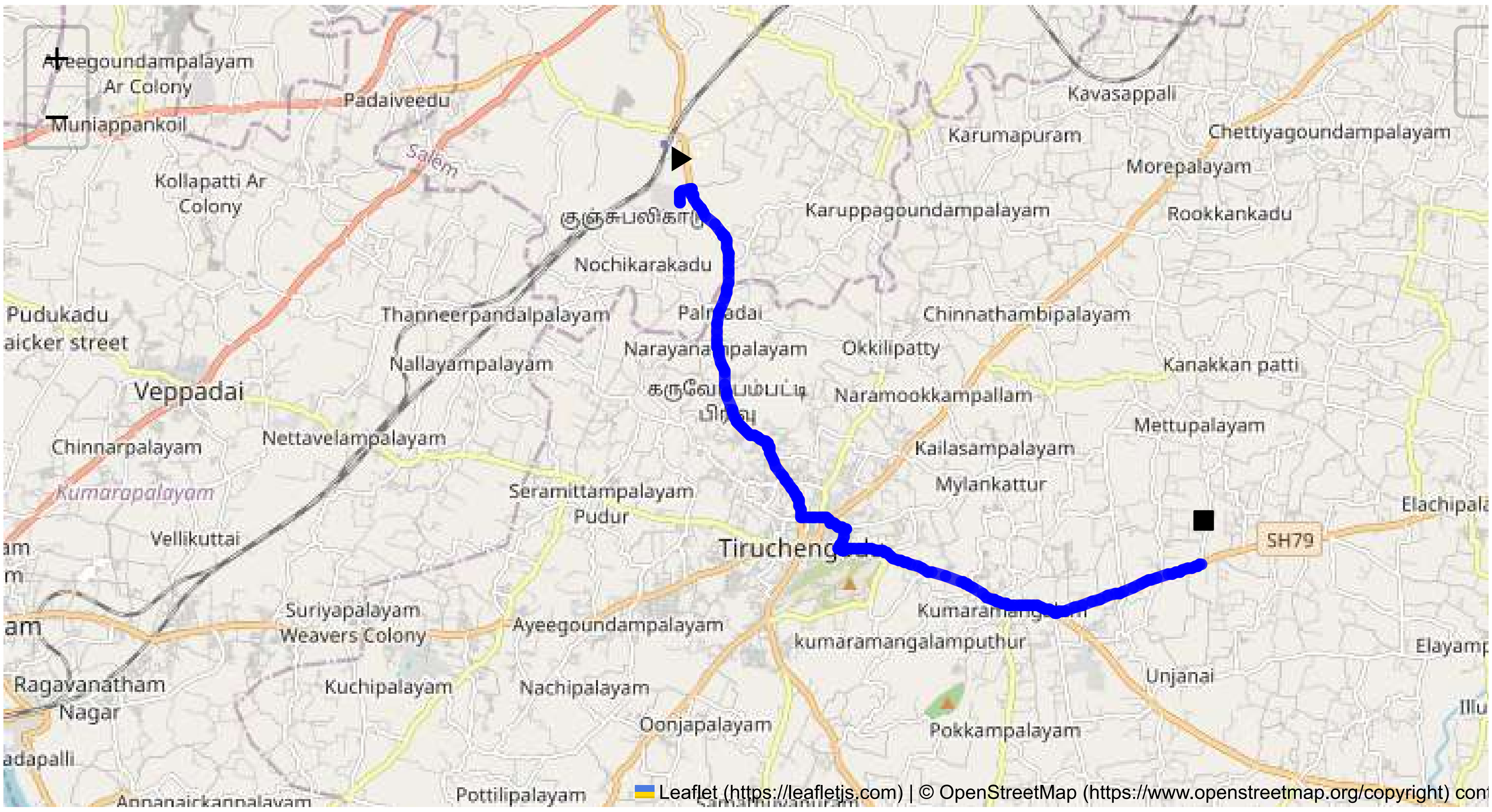
Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

Route Summary:
Total Distance: 16.17 km
Estimated Duration: 0.5 hours
Adjusted Duration (Heavy Vehicle): 0.6 hours
Start: (11.4381, 77.8734)
End: (11.375565, 77.965143)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Kavundampalaiyam covers approximately 16.17 kilometers, traversing various small towns and rural areas in Tamil Nadu. The journey starts at CVQF+23W in Sangagiri and takes a path through Puthur, Palmadai via SH 86, and Thondikaradu, before arriving at Kavundampalaiyam. It is essential to follow the highway markers and local signs for accurate navigation.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu typically experiences tropical weather with high humidity. The monsoon season is between October and December, which can lead to heavy rains and potential flooding. Travelers should be cautious of waterlogging and slippery roads during this period. In summer (March to June), high temperatures can lead to road surface softening and reduced tire traction.

3. Analysis of Traffic Patterns: The route is moderately traveled with instances of congestion likely around small-town centers during peak hours (8 AM - 10 AM and 5 PM - 7 PM). Specific congestion-prone areas include the markets and school zones in Tiruchengode near the Girls High School.

4. Assessment of Road Quality and Infrastructure: Overall, the road conditions are mixed, with state highways generally well-maintained. However, in some rural stretches, narrower roads, uneven surfaces, and occasional potholes may pose risks. Road infrastructure includes basic signage and lighting may be inadequate in less populated zones.

5. Suggestions for Alternative Routes for Emergencies: In emergencies, consider taking state highways such as SH 86 to bypass traffic-heavy areas. Having local maps and real-time GPS can help in re-routing efficiently in case of roadblocks or accidents.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Tamil Nadu follows national transportation safety standards which require proper documentation, clear labeling of hazardous materials, and adherence to speed limits. There are restrictions on the movement of hazardous materials during night hours in specific zones.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: Although specific local incident records are limited, general risks include vehicular accidents due to narrow roads and heavy traffic intersections. Occasional reports of overturning in curves or during rainy conditions in rural areas have been noted.

8. Environmental Considerations and Sensitive Areas: The route passes through agricultural zones where environmental sensitivity is high. Avoid chemical spills and ensure waste management practices are in place to protect these areas. Awareness of local wildlife crossings is necessary.

9. Analysis of Communication Coverage: While major towns along the route have good mobile network coverage, some rural stretches may experience weak signals or dead zones. It is advisable to have multiple means of communication (different network providers) and carry emergency contact numbers.

10. Estimated Emergency Response Times for Different Route Segments: Emergency response times can vary significantly based on location. In more populated areas like Tiruchengode, response times are relatively shorter (around 20-30 minutes), whereas remote sections like Palmadai may see delays up to 45-60 minutes due to distance from major emergency services.

12. Overall Summary of Risk Assessment: The route offers a medium risk level for transporting hazardous materials, driven by road quality variations, moderate traffic congestion, and potential weather impacts. Proper preparation, adherence to regulations, and awareness of alternative pathways and communication channels are vital to minimize risks. Regular vehicle maintenance checks, real-time weather updates, and driver training in emergency response can enhance safety during transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.38346, 77.89975	15 KM/Hr
6	Turn	Medium	11.38188, 77.90257	30 KM/Hr
7	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38346, 77.89975

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38188, 77.90257

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140